Project Pipeline

Pharmaceutical Operations Intelligence System

Step-by-step execution plan: 6 weeks

How to Use This Pipeline

Work through the phases in order. Tick each box when a step is done. Each phase ends with an **output** that must exist before you move on. Step numbers (for example 4.7) match the phase number, so you can refer to them in meetings and in the final report.

6-week plan

Week	Focus	Phases	Milestone
1	Set up, understand the data, ingest, profile, start cleaning	0, 1, 2, 3, start 4	Scope and data confirmed; data-quality report
2	Finish cleaning, integrate tables, start EDA	4, 5, start 6	Clean, integrated dataset
3	Finish EDA, features, splits, inventory baselines	6, 7, 8, start 9	EDA report; model-ready feature tables
4	Inventory models (MVP), quality models, first dashboard pages	9, 10, start 13	**MVP:** forecasts and stock-out and expiry alerts
5	Maintenance models, back-test, risk index, dashboard	9, 10, 11, 12, 13	All modules validated; dashboard demo
6	Testing, documentation, report, presentation, hand-over	14	Final submission

Suggested work split for the group

Adjust to the size of your group. Everyone reviews the cleaning rules and the final report.

Workstream	Covers
Data engineering	Phases 0–5: setup, ingestion, profiling, cleaning, integration, database
Inventory and demand	Phases 6–9 (inventory part): EDA, features, forecasting, stock-out and expiry models
Quality and maintenance	Phases 6–9 (quality and equipment part): control charts, classifiers, reliability
Dashboard, risk and reporting	Phases 11–14: back-test, risk index, Streamlit dashboard, documentation

Phase 0: Project Setup

Timing: Week 1 **Output:** a runnable project skeleton in Git

- ☐ **0.1** Create the Git repo with folders: data/raw, data/interim, data/processed, notebooks, src, models, reports, dashboard, tests.
- ☐ **0.2** Create the environment file (requirements.txt) and fix random seeds.
- ☐ **0.3** Treat raw data as read-only. Never edit it.
- ☐ **0.4** Create a config file for paths, column names, look-ahead days, service-level z value and risk weights.
- ☐ **0.5** Set up logging and a README. Agree Git rules: one branch per member, merge by pull request.

Phase 1: Business and Data Understanding

Timing: Week 1 **Output:** requirements log, data inventory, baseline KPIs

- ☐ **1.1** Confirm each stakeholder's decisions and KPIs (proposal Section 5.3).
- ☐ **1.2** List every data source with owner, format, date range and refresh rate.
- ☐ **1.3** Confirm which of the three domains you actually have: inventory and sales, quality, equipment.
- ☐ **1.4** If a domain is missing, pick open pharma sales data and build the synthetic data generator. Publish its assumptions and label every synthetic result.
- ☐ **1.5** Finalise the data dictionary, including the ★ fields (transaction date, lead time, temperature logs and CAPA status, maintenance type).
- ☐ **1.6** Calculate baseline KPIs: stock-out rate, fill rate, expiry write-off value, turnover, compliance rate, unplanned downtime.
- ☐ **1.7** Confirm the indicative targets against those baselines.

Phase 2: Data Ingestion

Timing: Week 1 **Output:** a repeatable load script and an ingestion log

- ☐ **2.1** Load each file (CSV or Excel) with explicit data types and encoding.
- ☐ **2.2** Validate the schema: expected columns present, no unexpected ones.
- ☐ **2.3** Record row counts, file hashes and load time.
- ☐ **2.4** Standardise column names: lowercase, snake_case, no spaces.
- ☐ **2.5** Save a raw snapshot to data/raw and a working copy to data/interim.

Phase 3: Data Profiling

Timing: Week 1 **Output:** a data-quality report listing every issue found

- ☐ **3.1** Check shape, data types, memory use and unique counts per column.
- ☐ **3.2** Count missing values per column and per row. Look for fake blanks: 0, -999, N/A, unknown, empty text.

- ☐ **3.3** Count exact and near-duplicate rows.
- ☐ **3.4** Produce summary statistics: min, max, mean, quartiles, skew.
- ☐ **3.5** Check value ranges and category lists.
- ☐ **3.6** Check date coverage, gaps and date order.
- ☐ **3.7** Check key coverage between tables: drug, batch, branch and equipment IDs.
- ☐ **3.8** Test stock reconciliation: opening + purchases − sales − write-offs = closing.

Phase 4: Data Cleaning

Timing: Weeks 1–2 **Output:** cleaned tables in data/interim, a cleaning log, unit tests for the rules

All domains

- ☐ **4.1** **Convert data types:** dates to datetime, text numbers to numeric (remove commas and currency signs), IDs to text, categories to category type.
- ☐ **4.2** **Standardise text:** trim spaces, fix case, correct spelling variants, merge synonyms.
- ☐ **4.3** **Standardise units and currency.** Document each conversion.
- ☐ **4.4** **Remove exact duplicates.** Resolve near-duplicates with a written rule (keep latest or most complete).
- ☐ **4.5** **Standardise keys** across tables: drug, batch, branch, equipment IDs.
- ☐ **4.6** **Handle missing values** by column, using a written rule: drop only if useless; impute only where defensible (median, mode or group value); add a missing-flag column where the gap means something. Never treat missing sales days as zero demand.
- ☐ **4.7** **Handle outliers:** detect with IQR, z-score and domain limits. Fix, cap or keep each one and record why.
- ☐ **4.8** **Log every change:** number of rows or cells affected and the reason.

Inventory and sales

- ☐ **4.9** Fix or remove negative quantities, negative prices and future-dated sales.
- ☐ **4.10** Check selling price against unit cost and flag oddities.
- ☐ **4.11** Flag zero-stock days as stock-out days. Demand is censored on those days.
- ☐ **4.12** Separate returns, write-offs and transfers from sales.
- ☐ **4.13** Complete the date index per product and branch (weekly).

Quality and compliance

- ☐ **4.14** Check that expiry date is after manufacture date.
- ☐ **4.15** Check audit scores are within the valid range.
- ☐ **4.16** Standardise deviation types and severity levels.
- ☐ **4.17** Recalculate CAPA status and ageing from dates.
- ☐ **4.18** Check temperature logs for sensor errors and flag excursions outside the product's limits (for example 2–8 °C).

Equipment and maintenance

- ☐ **4.19** Check that downtime hours are non-negative and plausible.
- ☐ **4.20** Make sure maintenance type is split into planned and corrective.

- ☐ **4.21** Check that failure dates fall after the asset's install date.
- ☐ **4.22** Keep assets with no failure and mark them as right-censored for survival analysis.

Phase 5: Integration and Data Model

Timing: Week 2 **Output:** integrated dataset and data model diagram (Deliverable 1)

- ☐ **5.1** Design the star schema. **Facts:** sales, stock, batches, deviations, maintenance. **Dimensions:** drug, supplier, branch, equipment, date.
- ☐ **5.2** Create surrogate keys.
- ☐ **5.3** Join the tables and compare row counts before and after each join.
- ☐ **5.4** Enforce referential integrity and report orphan records.
- ☐ **5.5** Load into PostgreSQL. If no database is available, use Parquet files.
- ☐ **5.6** Re-run the stock reconciliation on the integrated data.

Phase 6: Exploratory Data Analysis

Timing: Weeks 2–3 **Output:** EDA report that answers the research questions (Deliverable 2)

- ☐ **6.1** **Univariate:** distributions of demand, price, stock, audit score and downtime; counts for categories.
- ☐ **6.2** **Time series:** trend, seasonality, weekday and holiday effects, decomposition, autocorrelation.
- ☐ **6.3** **Demand segmentation:** ABC (value), XYZ (variability) and intermittent-demand classes.
- ☐ **6.4** **Bivariate:** demand against price, season, branch and supplier. Correlations (Pearson and Spearman) with a heatmap.
- ☐ **6.5** **Inventory:** turnover, days of cover, overstock, slow movers, expiry profile by batch.
- ☐ **6.6** **Supplier:** lead-time average and variation, fill rate, reliability.
- ☐ **6.7** **Quality:** Pareto of deviation causes, deviation and audit trends by branch, supplier and category.
- ☐ **6.8** **Equipment:** failure frequency by type, repeat failures, downtime distribution, planned versus corrective share.
- ☐ **6.9** **Cross-domain:** do missed maintenance and temperature excursions come before stock loss or deviations?
- ☐ **6.10** Answer each research question in proposal Section 4 with a chart and a sentence of evidence.

Phase 7: Feature Engineering

Timing: Week 3 **Output:** model-ready feature tables and a feature dictionary

Demand

- ☐ **7.1** Lags, rolling mean, rolling standard deviation and growth rates.
- ☐ **7.2** Calendar features: week, month, holidays, month-end, season.
- ☐ **7.3** Price features and the stock-out flag.

Inventory risk

- ☐ **7.4** Stock cover = current stock ÷ forecast daily demand.
- ☐ **7.5** Remaining shelf life and expiry-risk ratio.
- ☐ **7.6** Supplier reliability score and lead-time variability.

Quality and maintenance

- ☐ **7.7** Deviation rate, deviation severity score, audit-score trend, CAPA ageing, days since last audit.
- ☐ **7.8** Days since last maintenance, maintenance compliance index, past failure count, asset age, average downtime, temperature variance.

Finishing

- ☐ **7.9** Encode categories (one-hot or target encoding), fitted on training data only.
- ☐ **7.10** Scale only if the model needs it, with scalers fitted on training data only.
- ☐ **7.11** Build the target labels: stock-out within 14 days, expiry before sale, deviation or low audit next period, failure within the chosen horizon.

Phase 8: Data Splitting and Leakage Control

> **Timing:** Week 3 **Output:** train, validation and hold-out sets that are safe to use

- ☐ **8.1** Use time-based splits and rolling-origin cross-validation. Never use random splits.
- ☐ **8.2** Fit every imputer, scaler and encoder on training data only.
- ☐ **8.3** Compute each feature only from information available at prediction time.
- ☐ **8.4** Keep a final hold-out period untouched until the end.

Phase 9: Modelling

> **Timing:** Weeks 3–5 (inventory first, then quality, then maintenance) **Output:** trained, versioned models (Deliverables 3, 4, 5)

Inventory and demand (weeks 3–4)

- ☐ **9.1** Build baselines: seasonal-naive and moving average.
- ☐ **9.2** Train Prophet or SARIMA, and XGBoost. Use Croston-type methods for intermittent items.
- ☐ **9.3** Choose the best model per demand segment.
- ☐ **9.4** Calculate safety stock = $z \times \sigma(\text{daily demand}) \times \sqrt{(\text{lead time})}$ and reorder point = average daily demand × lead time + safety stock.
- ☐ **9.5** Train the stock-out classifier (14-day look-ahead).
- ☐ **9.6** Calculate the expiry-risk score: at risk if days of cover > days to expiry.

Quality and compliance (week 4)

- ☐ **9.7** Build baselines: majority class and logistic regression.
- ☐ **9.8** Train a decision tree and Random Forest.
- ☐ **9.9** Draw control charts on audit scores, deviation rates and temperature.

Maintenance (week 5)

- ☐ **9.10** Calculate MTBF, MTTR and availability = MTBF ÷ (MTBF + MTTR) per asset.
- ☐ **9.11** Train the failure classifier. Add a Weibull survival model if there are enough failures.
- ☐ **9.12** Rank assets by failure risk and criticality to produce the maintenance list.

For every classifier

- ☐ **9.13** Handle class imbalance with class weights, or resampling on training data only.
- ☐ **9.14** Tune hyperparameters with time-aware cross-validation.
- ☐ **9.15** Pick the decision threshold from the precision-recall curve and business cost.
- ☐ **9.16** Calibrate the probabilities. The risk index treats them as real probabilities.
- ☐ **9.17** Save and version each model.

Phase 10: Model Evaluation

Timing: Weeks 4–5 **Output:** evaluation report

- ☐ **10.1** **Forecasts:** WAPE, RMSE, MAE, R^2 and bias on the hold-out, compared with the baseline.
- ☐ **10.2** **Classifiers:** recall, precision, F1, PR-AUC and confusion matrix. Check them against the targets in Appendix B.
- ☐ **10.3** Analyse errors by segment, branch, product and season.
- ☐ **10.4** Explain results: feature importance, SHAP values or coefficients, with a plain-language driver for each flag.
- ☐ **10.5** Check stability across different time periods.
- ☐ **10.6** Adopt a model only if it beats its baseline by a meaningful margin.

Phase 11: Business Back-Test

Timing: Week 5 **Output:** back-test results against the KPI targets

- ☐ **11.1** Replay history under the recommended policies: reorder points, expiry actions, maintenance priorities.
- ☐ **11.2** Compare with what actually happened for stock-out days, expiry write-off value, turnover and unplanned downtime.
- ☐ **11.3** Run sensitivity checks on service level, look-ahead period and thresholds.
- ☐ **11.4** Report results as **simulated** savings and state the assumptions.

Phase 12: Integrated Risk Index

Timing: Week 5 **Output:** risk index and value-at-risk table per branch and product

- ☐ **12.1** Scale each risk (stock-out, expiry, compliance, equipment failure) to 0–1.
- ☐ **12.2** Agree the weights with stakeholders and document them.
- ☐ **12.3** Calculate the operational risk index as the weighted sum.
- ☐ **12.4** Calculate value at risk = stock value × risk probability.
- ☐ **12.5** Rank by value at risk and test how the ranking changes with different weights.

- ☐ **12.6** Attach a recommended action to every alert.

Phase 13: Dashboard and Deployment

> **Timing:** Weeks 4–6 (build pages as each module finishes) **Output:** working dashboard (Deliverable 6)

- ☐ **13.1** Create the scoring job that refreshes predictions in batches.
- ☐ **13.2** Build the Streamlit pages: executive overview, inventory and demand, expiry and procurement, quality and compliance, equipment health.
- ☐ **13.3** Add filters (branch, product, date), KPI cards, alert tables and CSV export.
- ☐ **13.4** Show the drivers next to every risk flag.
- ☐ **13.5** Show the data refresh date and model version on each page.
- ☐ **13.6** Test with a few sample users and fix usability issues.

Phase 14: Testing, Documentation and Hand-over

> **Timing:** Week 6 **Output:** final report, presentation and repository (Deliverables 7 and 8)

- ☐ **14.1** Write unit tests for cleaning rules, features and risk formulas.
- ☐ **14.2** Run the whole pipeline end to end, from raw files to the dashboard.
- ☐ **14.3** Make the repo runnable from the README in a few steps.
- ☐ **14.4** Write the final report: method, results, limitations, back-test, recommendations, ethics and privacy (no patient data; Kenya Data Protection Act, 2019).
- ☐ **14.5** Prepare the presentation and a short model card for each model.
- ☐ **14.6** Review every deliverable against its acceptance criteria.
- ☐ **14.7** Tag a release in Git and submit.

Appendix A: Key Formulas

Item	Formula
Safety stock	z × σ(daily demand) × √(lead time in days)
Reorder point	(Average daily demand × lead time) + safety stock
Days of cover	Current stock ÷ forecast daily demand. At risk if days of cover > days to expiry.
Availability	MTBF ÷ (MTBF + MTTR)
WAPE	Σ |actual − forecast| ÷ Σ actual
Value at risk	Stock value × probability of loss
Inventory turnover	Cost of goods sold ÷ average inventory

Appendix B: Success Targets

These are indicative. Confirm them against your baselines in Phase 1.

Area	Target
Demand forecasting	WAPE at least 15% better than seasonal-naive on the hold-out period
Stock-out prediction	Recall ≥ 0.80 with precision ≥ 0.60
Expiry-risk prediction	Recall ≥ 0.85 for high-value batches
Compliance-risk classification	Beats majority-class and logistic baselines; drivers are explainable
Equipment-failure prediction	Recall ≥ 0.75 with enough warning to schedule work
Back-test: stock-out days	≥ 20% reduction
Back-test: expiry write-offs	≥ 25% reduction
Back-test: unplanned downtime	≥ 15% reduction

Appendix C: Deliverables Checklist

#	Deliverable	Due week	Phase
1	Clean, integrated dataset and data dictionary	2	1–5
2	Exploratory data analysis report	3	6
3	Inventory and demand optimisation model	4	7–10
4	Quality compliance monitoring framework	4–5	9–10
5	Preventive maintenance analytics module	5	9–10
6	Interactive dashboard	5–6	12–13
7	Final technical report and presentation	6	11, 14
8	Source code repository	6	0, 14